

Titanic Dataset Analysis - Project Report

Data Analysis Project - Week 4, Data Science with Python

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Objective

The goal of this project was to explore the Titanic passenger dataset and understand the factors associated with whether a passenger survived the disaster. I focused on gender, passenger class, age, fare, and port of embarkation. Python libraries such as Pandas, Matplotlib, and Seaborn were used to clean the data, perform analysis, and create visualizations to identify important patterns.

About the Dataset

The dataset contains records for **891 passengers across 12 columns**, including PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked.

Before performing the analysis, I checked for missing values. The Age column had **177 missing entries**, Embarked had **2**, and Cabin had **687** missing entries, which is almost 77% of the column.

Data Cleaning

Since Age is an important variable for this analysis, I filled the missing Age values using the **median age** instead of removing those rows. The two missing Embarked values were filled using the **most common port (mode)**.

The Cabin column contained a very large number of missing values, so it was removed from the analysis. After these cleaning steps, there were no missing values in the remaining columns used for analysis.

Key Findings

1. Overall Survival

Out of 891 passengers, **342 survived and 549 did not survive**, giving an overall survival rate of approximately **38.4%**.

2. Gender and Survival

Gender showed the largest difference in survival rate in this analysis. Female passengers had a survival rate of approximately **74.2%**, while male passengers had a survival rate of approximately **18.9%**.

This shows a clear difference in survival outcomes between male and female passengers. The historical practice of prioritizing women and children during evacuation provides additional context for this pattern.

3. Passenger Class and Survival

Passenger class was also strongly associated with survival.

Group	Survival Rate
Female	74.2%
Male	18.9%
1st Class	63.0%
2nd Class	47.3%
3rd Class	24.2%

Passengers in 1st class had a higher survival rate than passengers in 2nd and 3rd class. The analysis demonstrates a clear relationship between passenger class and survival, although the dataset alone does not establish the exact reasons for this difference.

4. Age

Most passengers were between approximately 20 and 40 years old. The age distribution also differed between survivors and non-survivors. Younger passengers, including children, showed different survival patterns compared with older passengers.

5. Fare

Passengers who survived generally had higher fare values than passengers who did not survive. This pattern is consistent with the relationship between fare and passenger class, as higher fares were generally associated with higher passenger classes.

6. Embarkation Port

Passengers who boarded at **Cherbourg (C)** showed a higher survival rate compared with passengers who boarded at **Southampton (S)** or **Queenstown (Q)**.

This difference may partly reflect differences in the passenger groups and classes represented at each port. The analysis shows an association between embarkation port and survival, but it does not establish that the port itself caused the difference.

7. Correlation

The correlation heatmap showed a **negative correlation between Survived and Pclass**. Since 1 represents 1st class and 3 represents 3rd class, this indicates that survival tended to decrease as the Pclass value increased.

Fare showed a **positive correlation with Survived**, which is consistent with the higher survival rates observed among passengers with higher fares and higher passenger classes.

Conclusion

This analysis shows that survival outcomes in the Titanic dataset were strongly associated with **gender and passenger class**, with additional patterns related to age, fare, and embarkation port.

Female passengers and passengers travelling in higher classes had higher observed survival rates in this dataset. Age, fare, and embarkation port also showed differences between survivors and non-survivors.

This project helped me practice the complete data analysis workflow, including loading a real-world dataset, checking and cleaning missing data, performing exploratory analysis, and using visualizations to identify meaningful patterns that are difficult to observe from raw data alone.

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab